

YZV 447E TERM PROJECT REPORT
Fall 2025
Report 1: Hypothesis and Experiment Design

Team Name: Axion

Research question that motivated this experiment

“Bruce and Young (1986) proposed that face recognition involves several independent sub-processes. [...] The processing of facial expression is assumed to be independent of the processing of facial identity. This view is supported by clinical evidence showing that patients can have deficits in recognizing identity while retaining the ability to recognize expressions, and vice versa.”

The Student’s Guide to Cognitive Neuroscience, 4th Ed. Jamie Ward, Chapter 7

Based on this classic model, we question whether strong emotional cues (like anger or happiness) interfere with the processing of invariant facial features such as gender, or if the functional independence holds true even under high-arousal conditions.

Important aspects to study

While the cognitive model suggests independence, evolutionary psychology suggests that "threat" signals (Angry faces) grab attention automatically. Testing whether this evolutionary mechanism overrides the cognitive "independence" of face processing routes is a crucial aspect of this study. If the classic model is robust, even "Angry" faces shouldn't delay gender recognition.

Hypothesis

Processing of facial expression and facial identity are functionally independent. (Because of this, the response time for gender decisions will be the same regardless of whether the face is Happy, Angry, or Neutral. We do not expect significant differences in RT across emotional conditions).

Design parameters

- Type(s) of stimuli: Human face photographs selected from the KDEF (Karolinska Directed Emotional Faces) dataset.
- Experimental paradigm: Gender Decision Task (Implicit Emotion Processing).
- Conditions 2x3 Design
 - Category 1: Expression (Happy, Angry, Neutral)
 - Category 2: Gender (Male, Female)
- Some experimental details: Number of Trials: 40 trials per condition (Total 240 trials). ISI (Inter-stimulus Interval): 500 ms fixation cross. Trial duration: 1000 ms (or until response).
- Dependent variables: Reaction Time (RT) and Accuracy (Acc).
- Question to ask the participant and collected response(s): In this experiment, the participant sees a face on the screen. The explicit task is to decide the gender of the person, ignoring the facial expression.

Press **F** if what you see is a **WOMAN**.

Press **M** if what you see is a **MAN**.

Eg:



(subject presses F)

Eg:



(subject presses M)

References

- Ward, J. (2019). The Student's Guide to Cognitive Neuroscience (4th ed.). Routledge.
- Bruce, V., & Young, A. (1986). Understanding face recognition. British Journal of Psychology, 77(3), 305–327.
- Lundqvist, D., Flykt, A., & Öhman, A. (1998). The Karolinska Directed Emotional Faces (KDEF) [Database of standardized facial images]. Psychology Section, Department of Clinical Neuroscience, Karolinska Institutet.

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